

Homework (Jalapeno)

- Q1.** Factor the expression: $x^2 + 5x + 4x + 20$
(A) $x(x + 5) + 4(x + 5)$
(B) $x(x + 4) + 5(x + 4)$
(C) $x^2 + 9x + 20$
(D) $x(x + 4) + 5(x + 4)$
(E) $x(x + 5) + 4(x + 5)$
- Q2.** Factor by grouping: $3x^2 + 12x + 4x + 16$
(A) $3x(x + 4) + 4(x + 4)$
(B) $3x(x + 4) + 4(x - 4)$
(C) $x(3x + 4) + 4(3x + 4)$
(D) $(3x + 4)(x + 4)$
(E) none of these
- Q3.** Which expression is NOT factored by grouping?
(A) $a(b + c)$
(B) $ab + ac + bc + bd$
(C) $ab + ac + ad + bc$
(D) $2x(x + 3) + 5(x + 3)$
(E) $3(x + 4) + 2(x + 4)$
- Q4.** The Code of Hammurabi was written around 1754 BCE. It included rules for the construction of homes. If the length of a home is l meters, the width is w meters, and the area of the home is given as $lw + 2l + 2w + 4$, what is the factored form?
(A) $l(w + 2) + 2(w + 2)$
(B) $lw + 2l + 2w + 2$
(C) $l(w + 2) + 2(lw + 2)$
(D) $2(l + w) + lw + 2$
(E) none of the above
- Q5.** The Rosetta Stone includes a decree from ancient Egypt. If a scribe had a pieces of papyrus and b writing utensils, and he received a new writing utensils and b new pieces of papyrus, his total items could be represented by the expression $ab + ac + bd + bc$. What is the factored form?
(A) $a(b + c) + b(b + c)$
(B) $a(b + c) + b(d + c)$
(C) $ab + ac + bd + bc$
(D) none of these
(E) $b(a + c) + c(a + b)$
- Q6.** Which of the following is an example of factoring by grouping?
(A) $6x + 3y + 2z$
(B) $x^2 + 4x + 3x + 12$
(C) $x^2 - 4x + 4$
(D) $6x - 3$
(E) $3x^2 + 2y$
- Q7.** The ancient city of Babylon had x temples, y gardens, x canals, and y bridges. The total number can be represented by $xy + xz + yz + yx$. What is the factored form?
(A) $x(y + z) + y(x + z)$
(B) $x(y + z) + y(z + x)$
(C) $x(y + z) + y(x + z) + zx$
(D) $xy + xz + yz + yx$
(E) $x(y + z) + z(y + x)$
- Q8.** The Great Pyramid of Giza required the use of x workers for y days to move z blocks. If each worker could move x blocks per day and there were y groups of workers, what is the factored expression of $xy + xz + yz + yx$?
(A) $x(y + z) + y(x + z)$
(B) $z(x + y) + x(y + z)$
(C) $x(y + z) + x(y + z)$
(D) $xy + xz + yz + yx$
(E) $y(x + z) + x(y + z)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $x^2 + 5x + 4x + 20$ and verify by multiplying. Provide all steps.
- Q10.** A pharaoh ordered the construction of a temple with x pillars and y obelisks. If the total number of structures is $xy + xz + yz + yx$, factor the expression and explain the reasoning behind factoring by grouping.

Chef's Tips

- Tip 1: Be sure to check for common factors before grouping
- Tip 2: Verify factoring by multiplying the expression
- Tip 3: Use the distributive property when factoring by grouping